Job Recommendation System

The **Job Recommendation System Using LinkedIn User Profiles** is a Machine Learning-based web application developed to help job seekers find suitable employment opportunities more easily. With the rapid growth of online professional platforms and the large number of available job postings, users often find it difficult to identify jobs that match their skills, qualifications, experience, and career interests. The proposed system addresses this problem by analyzing LinkedIn user profile information and available job descriptions to provide personalized job recommendations. The system considers details such as technical skills, education, work experience, certifications, projects, job title, industry, and professional interests to identify suitable job opportunities.

Traditional job portals may provide generic recommendations or depend mainly on keyword matching, which can result in irrelevant job suggestions. Users may also need to spend considerable time manually searching and filtering job postings. The proposed system improves this process by using Machine Learning and Natural Language Processing techniques to understand user profiles and job descriptions. It uses a hybrid recommendation approach that combines content-based and collaborative filtering concepts. This allows the system to consider both the information available in the user's profile and behavioral patterns when generating recommendations.

The system starts with user registration and authentication. A user provides details such as name, email, mobile number, skills, education, experience, certifications, projects, and job preferences. The administrator verifies and activates the registered account before the user can access the system. Users can update their professional information and upload profile or job-related datasets in the required format. The administrator can manage users, activate or deactivate accounts, manage datasets, and monitor system activities. This provides controlled access and helps maintain the reliability and security of the application.

After collecting the required information, the system performs data preprocessing on user profiles and job posting data. Raw data may contain missing values, inconsistent formats, or unnecessary information. Therefore, the system performs data cleaning, handles missing values, encodes categorical data, and processes textual information. Natural Language Processing techniques such as tokenization, stop-word removal, and **TF-IDF (Term Frequency–Inverse Document Frequency)** are used to convert textual information into numerical features. Feature engineering is also performed to identify useful attributes such as skill similarity, experience level, and job relevance.

The feature extraction process identifies important information from LinkedIn profiles and job descriptions, including skills, education, work experience, certifications, and job titles. TF-IDF is used to represent textual information numerically so that the system can compare user profiles with job requirements. These extracted features are then provided to the Machine Learning model. The project

uses the **Random Forest algorithm** for training and prediction. The model learns patterns from the processed dataset and predicts suitable job roles based on the user's skills and qualifications.

The recommendation module uses the prediction generated by the Machine Learning model to identify suitable jobs. It compares the user's profile with available job descriptions and ranks the matching jobs according to their relevance and similarity scores. The system can display information such as the recommended job title, company name, required skills, and matching score. This helps users quickly understand which opportunities are most relevant to their profile and reduces the time required for manual job searching.

The system follows a client-server architecture. The user interacts with a web-based frontend and provides profile information such as skills, education, and interests. This information is sent to the **Django backend**, where the TF-IDF Vectorizer converts textual information into numerical feature vectors. These features are passed to the trained Random Forest Machine Learning model, which identifies patterns between user profiles and job roles using the jobs dataset. The prediction module returns the recommended job to the frontend, while the visualization module provides additional information about relevant skills and job-related data. User inputs and prediction results can also be stored in the database for future reference.

The project is implemented using **Python, Django, HTML, CSS, JavaScript, SQLite, Natural Language Processing techniques, TF-IDF, and Random Forest**. Python is used for data processing and Machine Learning implementation, while Django provides the web application framework. HTML, CSS, and JavaScript are used to design the user interface, and SQLite is used for database management. The model performance is evaluated using metrics such as **accuracy, precision, recall, and F1-score**. A graphical dashboard can also display model performance, job trends, recommendation distribution, and feature importance.

The system provides benefits to both job seekers and organizations. For fresh graduates, it can suggest entry-level opportunities based on their educational background and skills. For experienced professionals, it can identify opportunities according to their existing expertise and experience. Organizations can also benefit by improving recruitment efficiency and identifying suitable candidates based on their skills and qualifications. The system reduces manual effort, saves time, improves job-candidate matching, and helps create a better connection between job seekers and employers.

The project can be further enhanced by integrating real-time job portals to provide continuously updated job opportunities. Advanced Machine Learning, Deep Learning, Transformer-based models, and improved NLP techniques can be used to understand the semantic meaning of skills and job descriptions more effectively. Other possible enhancements include mobile application support, personalized career guidance, multilingual support, improved security and privacy, employer

integration, AI chatbot assistance, and skill-gap analysis. These features can make the system more intelligent and useful for both job seekers and recruiters.

Overall, the **Job Recommendation System Using LinkedIn User Profiles** provides an intelligent approach to job searching by combining user profile analysis, NLP, feature extraction, and Machine Learning. By comparing a user's skills, qualifications, experience, and preferences with job requirements, the system generates relevant and personalized job recommendations. The project aims to reduce the time and effort involved in searching for suitable jobs while improving the quality of job matching. It demonstrates how Machine Learning can be effectively applied to recruitment and career recommendation systems to provide more efficient and personalized employment opportunities.